ETH zürich

D INFK

Probabilistic Artificial Intelligence

Gaussian Processes

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Recall: Bayesian Linear Regression

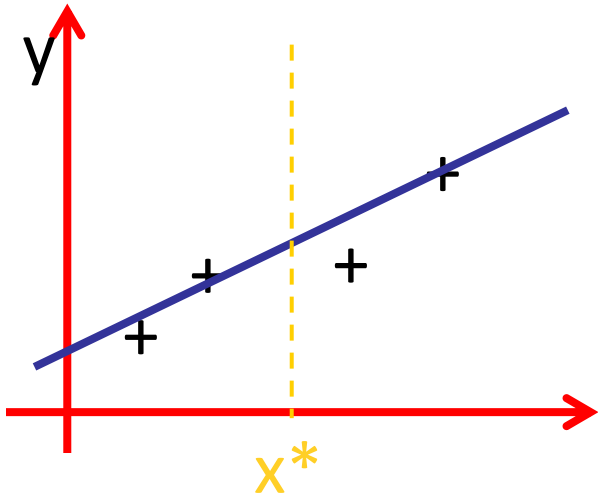
Prior: $p(w) = \mathcal{N}\left(w; 0, \sigma_p^2 I\right)$

Likelihood: $p(y \mid x, w) = \mathcal{N}(y; w^T x, \sigma_n^2)$

Posterior over **w**: $p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{y}) = \mathcal{N}(\boldsymbol{w}; \bar{\mu}, \bar{\Sigma})$

$$\bar{\mu} = \left(\boldsymbol{X}^T\boldsymbol{X} + \frac{\sigma_n^2}{\sigma_p^2}\boldsymbol{I}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

$$\bar{\Sigma} = \sigma_n^2\left(\boldsymbol{X}^T\boldsymbol{X} + \frac{\sigma_n^2}{\sigma_p^2}\boldsymbol{I}\right)^{-1}$$



For test point $\mathbf{x}^*$ define $f^* = \mathbf{w}^T\mathbf{x}^*$. Then:

$$p(f^* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}^*) = \mathcal{N}(\bar{\mu}^T\mathbf{x}^*, \mathbf{x}^{*T}\bar{\Sigma}\mathbf{x}^*)$$

and $$p(y^* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}^*) = \mathcal{N}(\bar{\mu}^T\mathbf{x}^*, \mathbf{x}^{*T}\bar{\Sigma}\mathbf{x}^* + \sigma_n^2)$$

Ridge regression vs Bayesian lin. regression

- Ridge regression: predict using *MAP estimate* for weights

$$\hat{\mathbf{w}} = \arg\max_{\mathbf{w}} p(\mathbf{w} \mid \mathbf{x}_{1:n}, y_{1:n})$$

$$p(y^* \mid \mathbf{x}^*, \hat{\mathbf{w}}) = \mathcal{N}(y^*; \hat{\mathbf{w}}^T \mathbf{x}^*, \sigma_n^2)$$

- BLR: predict by averaging *all* **w** acc. to posterior:

$$p(y^* \mid \mathbf{x}^*, \mathbf{x}_{1:n}, y_{1:n}) = \int p(y^* \mid \mathbf{x}^*, \mathbf{w}) p(\mathbf{w} \mid \mathbf{x}_{1:n}, y_{1:n}) d\mathbf{w}$$

$$= \mathcal{N}(y^*; \hat{\mathbf{w}}^T \mathbf{x}^*, \sigma_n^2 + \mathbf{x}^{*T} \bar{\Sigma} \mathbf{x}^*)$$

Aleatoric vs. epistemic uncertainty

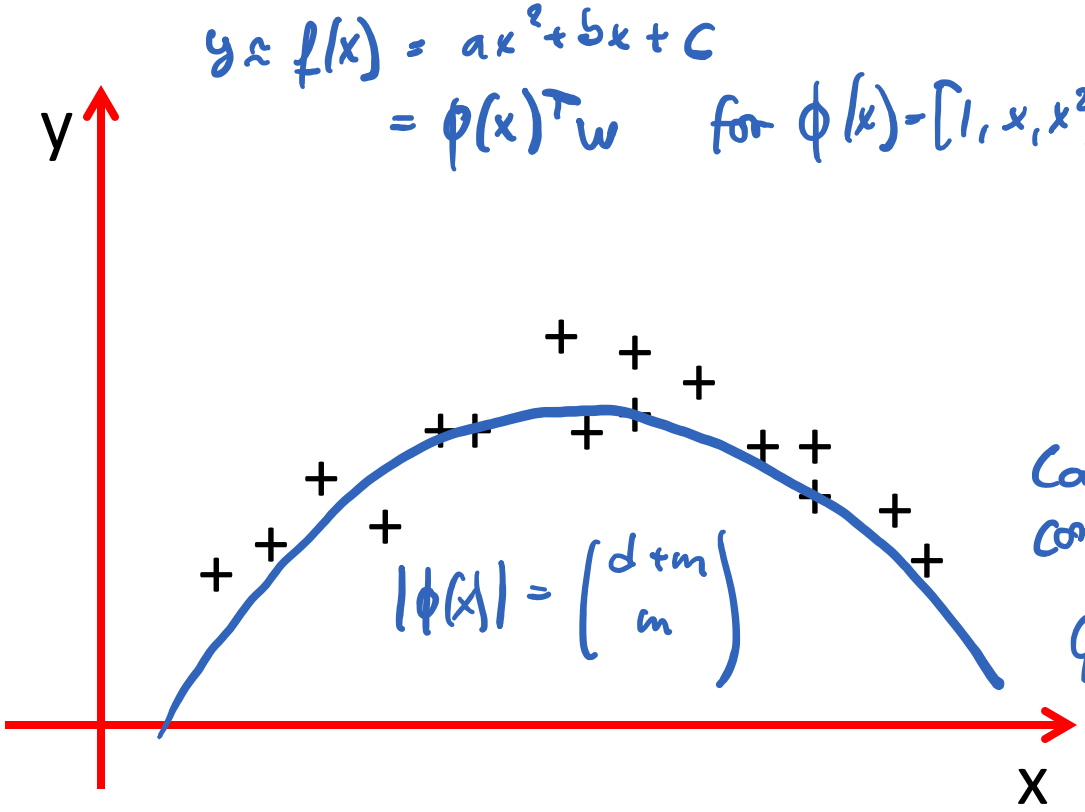
$$p(y^* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}^*) = \mathcal{N}(\bar{\mu}^T \mathbf{x}^*, \mathbf{x}^{*T} \bar{\Sigma} \mathbf{x}^* + \sigma_n^2)$$

Uncertainty about f* (epistemic)

Noise / uncertainty about y* given f* (aleatoric)

- Can distinguish two forms of uncertainty
  - Epistemic uncertainty: Uncertainty about the model due to the lack of data
  - Aleatoric uncertainty: Irreducible noise

What about nonlinear functions?



$$f(\mathbf{x}) = \sum_{i=1}^{d} w_i \phi_i(\mathbf{x})$$

Can generalize to d-dimensions by considering all monomials of degree m

$\phi(x) = [1, x_1 \dots x_d, x_1 x_2, \dots x_1 x_d, \dots x_{d-1} x_d, x_1^2 \dots x_d^2, \dots x_{d-m+1} \dots x_d^m]$

Recall: Can apply linear method (like BLR) on nonlinearly transformed data. However, computational cost increases with dimensionality of the feature space!

The „Kernel Trick“

- Express problem s.t. it only depends on inner products
- Replace inner products by kernels

Eg. $\phi(x) = [\text{all monomials up to degree } m]$

$\Rightarrow \phi(x)^T \phi(x') = (1 + x^T x')^m$

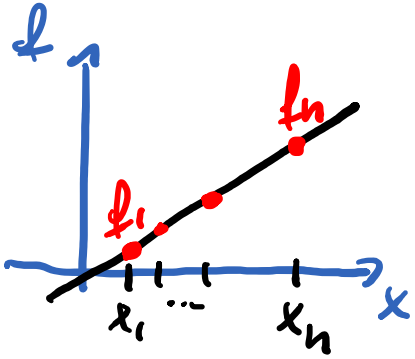
$$\mathbf{x}_i^T \mathbf{x}_j \quad \Rightarrow \quad k(\mathbf{x}_i, \mathbf{x}_j)$$

$\phi(x_i)^T \phi(x_j)$

How do we kernelize Bayesian linear regression?

Weight vs function space view

- Assume Gaussian distribution on the weights
- This implies Gaussian distribution on the predictions
- Suppose we consider an arbitrary (finite) set of inputs **X**
- The predictive distribution is given by



$f_{1:n} = f = [f_1 \dots f_n]^T$

$f_i = x_i^T w$

$f = X^T w$

$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

$w \sim \mathcal{N}(0, I)$

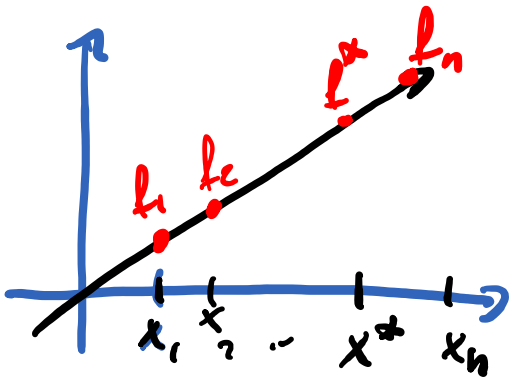
$f \sim \mathcal{N}(0, \underbrace{XX^T}_{K})$

$K = XX^T \in \mathbb{R}^{n \times n}$

$K_{ij} = x_i^T x_j \rightsquigarrow \underbrace{\phi(x_i)^T \phi(x_j)}_{k(x_i, x_j)}$

Predictions in "function space"

- Sps. we're given data **X**, **y**, and want to predict y* for x*



$$\tilde{X} = \begin{pmatrix} X \\ \hline x^{*T} \end{pmatrix}, \quad \tilde{y} = \begin{pmatrix} y \\ \hline y^* \end{pmatrix}, \quad \tilde{f} = \begin{pmatrix} f \\ \hline f^* \end{pmatrix}$$

$$\tilde{f} = \tilde{X} w, \quad \tilde{y} = \tilde{f} + \tilde{\varepsilon}$$

$$\Rightarrow \tilde{y} \sim \mathcal{N}(0, \underbrace{\tilde{X}\tilde{X}^T}_{K} + \sigma_n^2 \cdot I)$$

$$p(y^* | y_{1:n}) \equiv p(y^* | y_{1:n}, X_{1:n}, x^*) = \mathcal{N}(y^*; \mu_{x^*|X_{1:n},y_{1:n}}, \sigma^2_{x^*|X_{1:n}})$$

Key insight

- For prior $\mathbf{w} \sim \mathcal{N}(0, \mathbf{I})$, the predictive distribution over $\mathbf{f} = \mathbf{X}\mathbf{w}$ is Gaussian

$$\mathbf{f} \sim \mathcal{N}(0, \mathbf{X}\mathbf{X}^T)$$

- Thus, data points only enter as inner products!
- Can kernelize: $\mathbf{f} \sim \mathcal{N}(0, \mathbf{K})$ where

$$K_{\mathbf{x},\mathbf{x}'} = \phi(\mathbf{x})^T \phi(\mathbf{x}')$$

What about infinite domains?

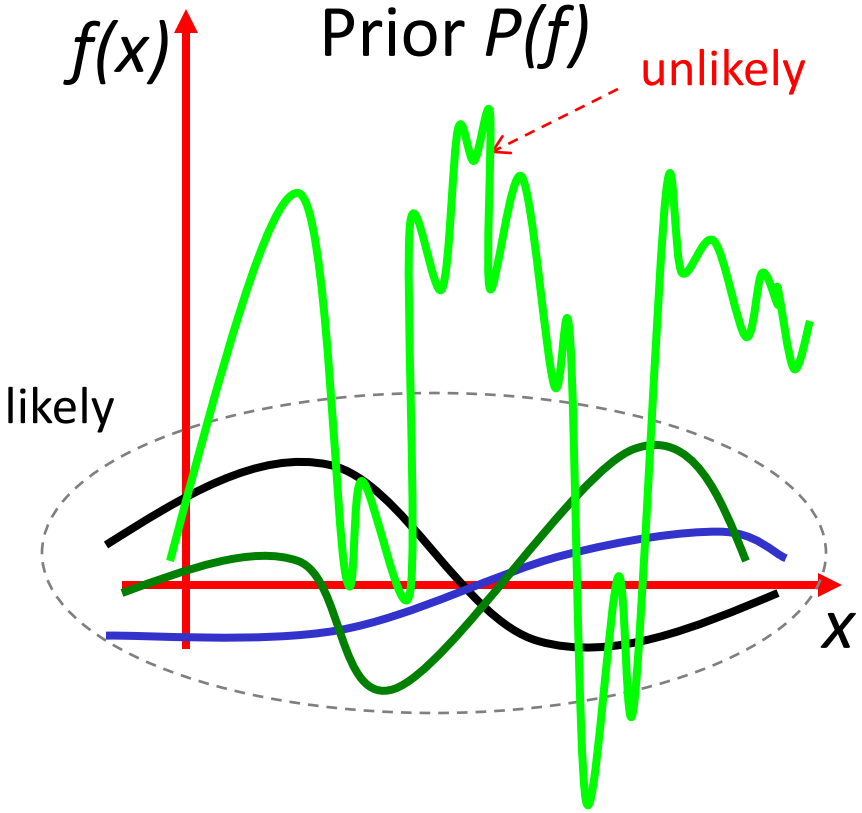
- The previous construction can be generalized to infinitely large domains X
- The resulting random function is called a Gaussian process

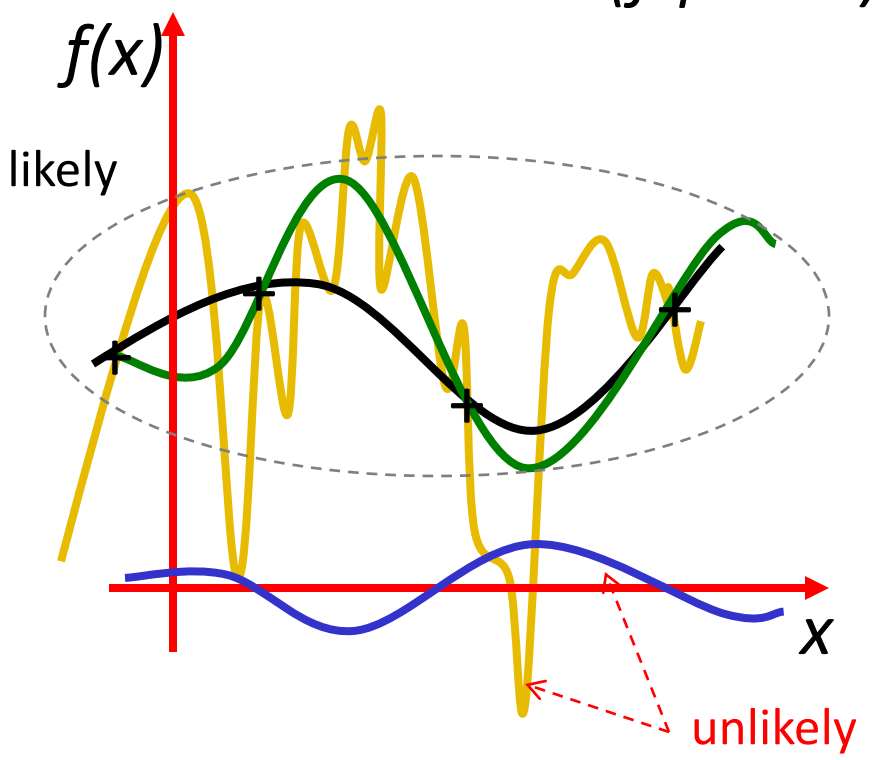
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*

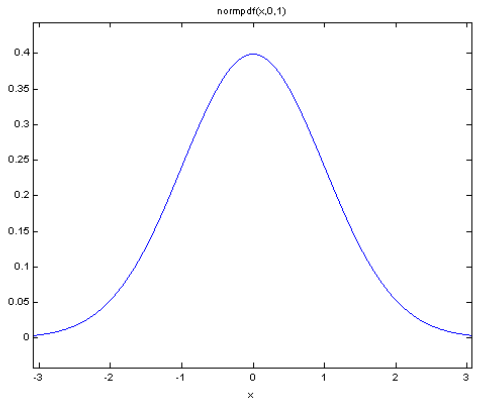
➔ Posterior: *P(f | data)*



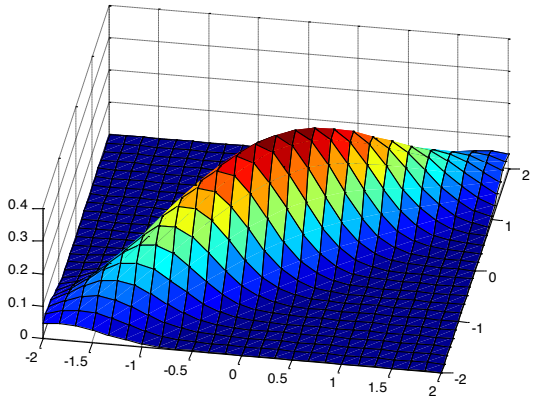


Predictive uncertainty + tractable inference

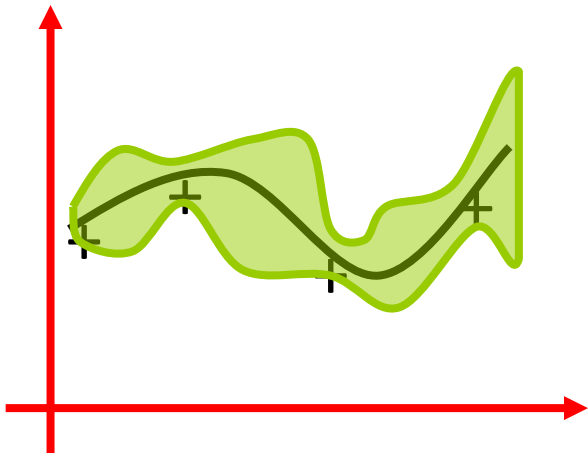
Gaussian Processes



Normal dist.
(1-D Gaussian)



Multivariate normal
(n-D Gaussian)



Gaussian process
(∞-D Gaussian)

- Gaussian process (GP) = normal distribution over *functions*
- Finite marginals are multivariate Gaussians
- Closed form formulae for Bayesian posterior update exist
- Parameterized by *covariance function k(x,x') = Cov(f(x),f(x'))*

Gaussian process

A Gaussian Process (GP) is an

(infinite) set of random variables, indexed by some set X

i.e., there exists functions $\mu : X \to \mathbb{R} \quad k : X \times X \to \mathbb{R}$

and for each $x \in X$ there's a random variable f_x

such that for all $A \subseteq X$, $A = \{x_1, \ldots, x_m\}$

it holds that $f_A = [f_{x_1}, \ldots, f_{x_m}] \sim N(\mu_A, K_{AA})$

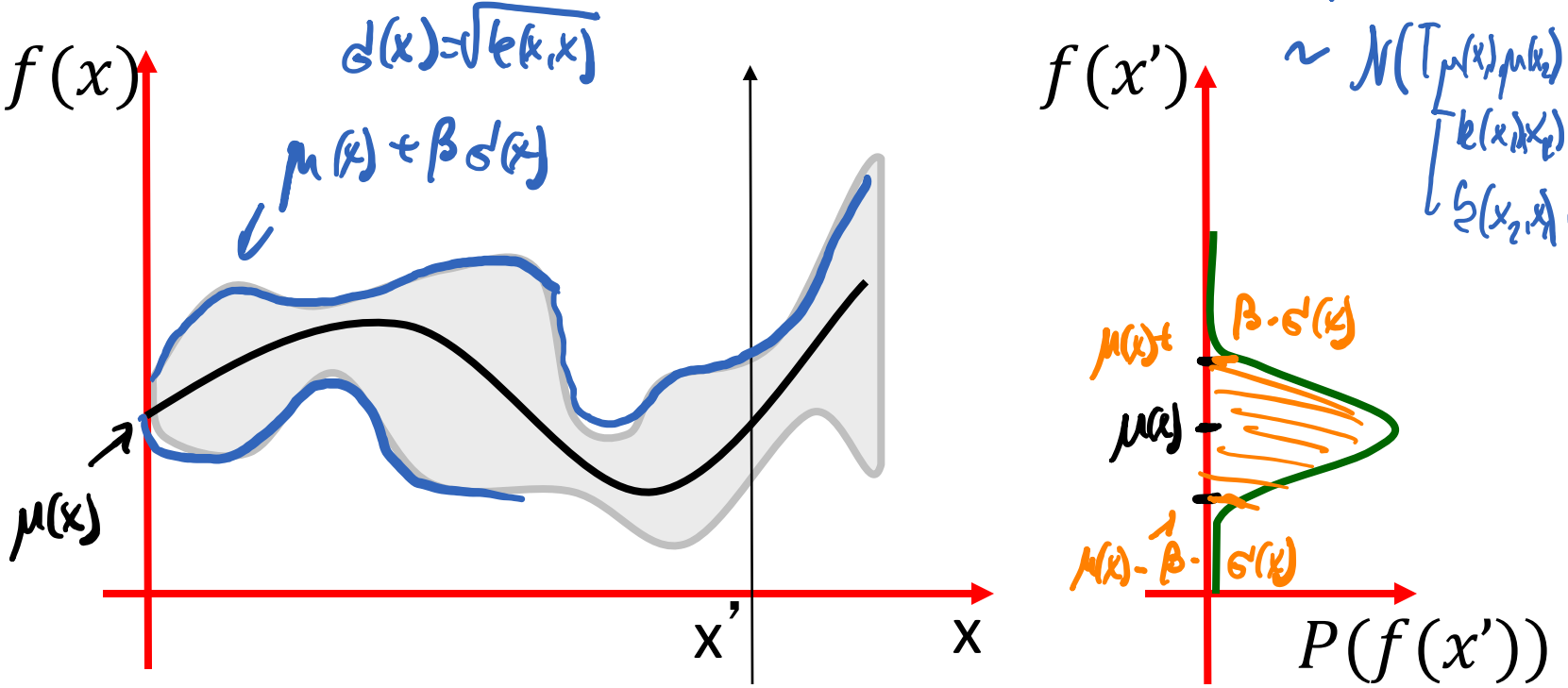
for $\mathbf{K}_{AA} = \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) & \ldots & k(x_1, x_m) \\ \vdots & & \vdots & \\ k(x_m, x_1) & k(x_m, x_2) & \ldots & k(x_m, x_m) \end{pmatrix}$ and $\mu_A = \begin{pmatrix} \mu(x_1) \\ \vdots \\ \mu(x_m) \end{pmatrix}$

k is called covariance (kernel) function

μ is called mean function

We write: $f \sim GP(\mu, k)$ and define $f(x) := f_x$

GP Marginals



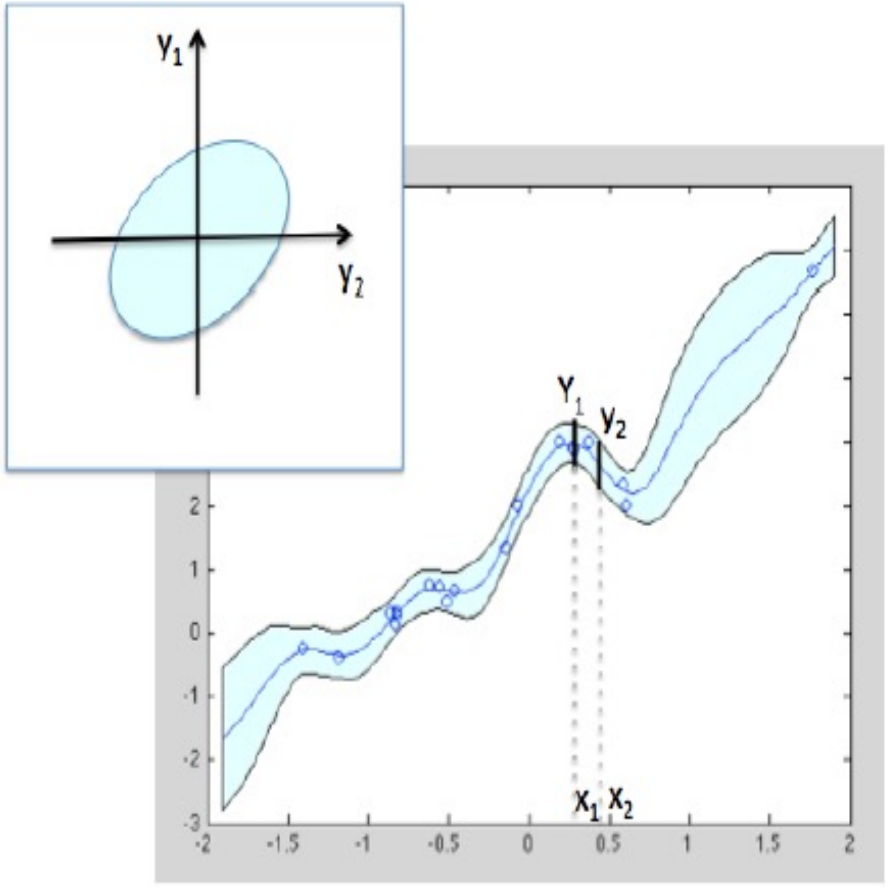
$[f(x_1), f(x_2)]$
Bivariate Gaussian
$\sim \mathcal{N}\left(\begin{bmatrix}\mu(x_1)\\ \mu(x_2)\end{bmatrix}, \begin{bmatrix}k(x_1,x_1) & k(x_1,x_2)\\ k(x_2,x_1) & k(x_2,x_2)\end{bmatrix}\right)$

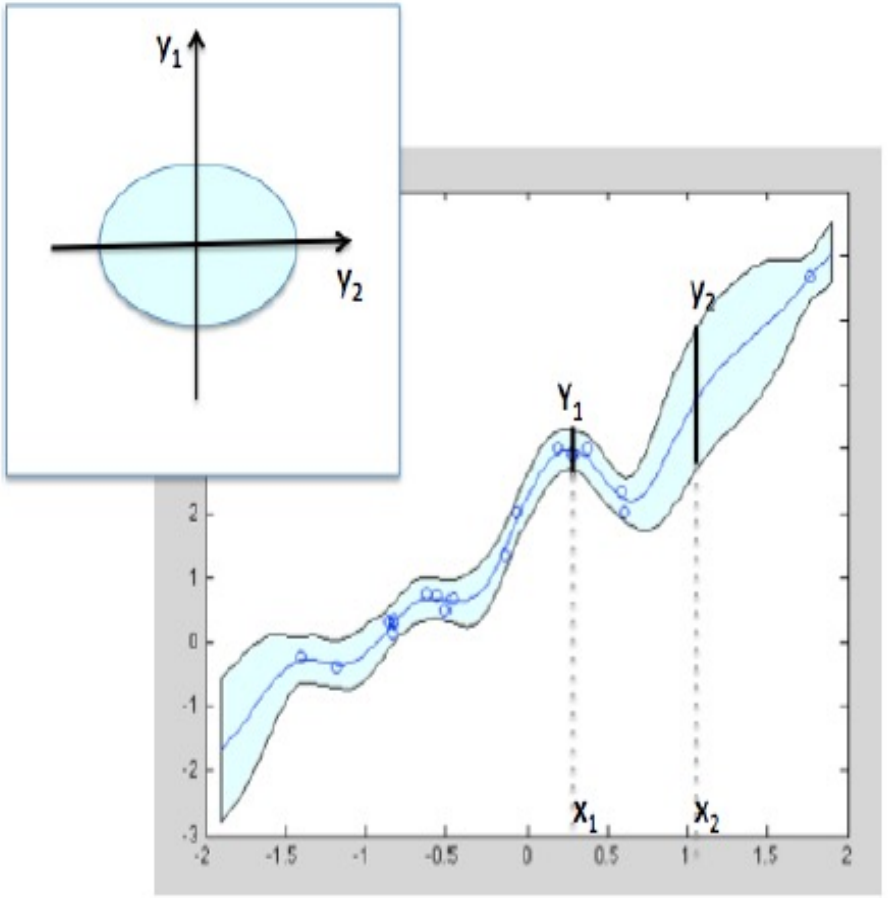
Typically, primarily interested in marginals, i.e.,

$$p(f(x)) = \mathcal{N}(f(x); \mu(x); k(x,x))$$

variance of $f(x) = \sigma^2(x)$

Illustration of covariance





Covariance (kernel) functions

- *k* must be symmetric

$$k(x, x') = k(x', x) \text{ for all } x, x'$$

- *k* must be positive semi-definite

For all *A*: K_{AA} is positive semi-definite matrix

Why? $f \sim N(0, K_{AA})$, $f \in \mathbb{R}^m$, $v \in \mathbb{R}^m$

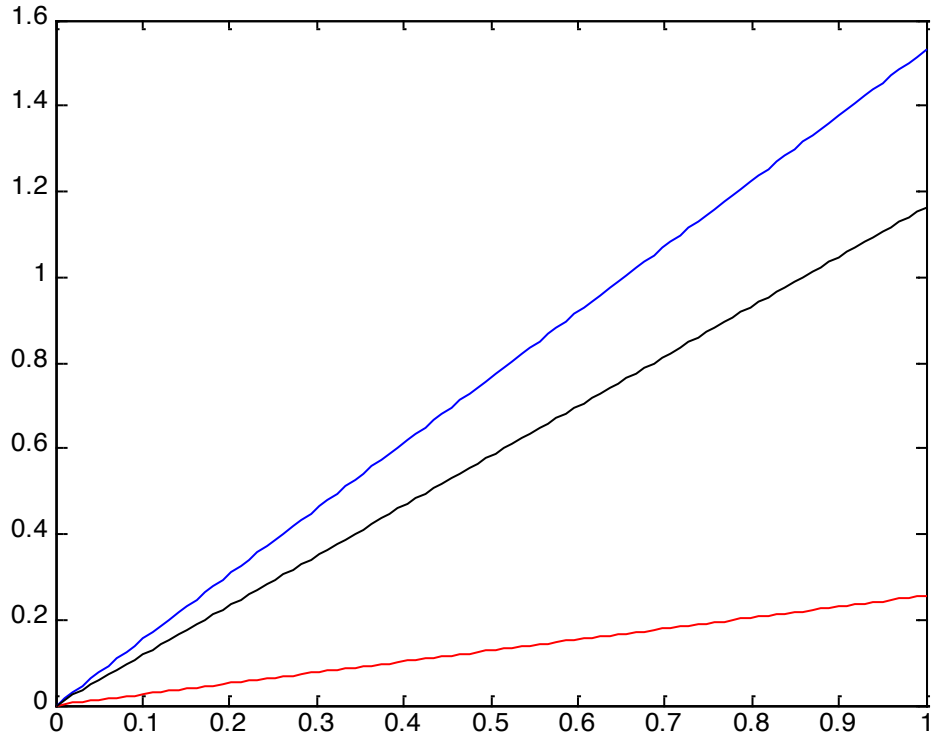
$v^T f \sim N(0, v^T K_{AA} v)$

$v^T K_{AA} v$: variance of $v^T f$ ≥ 0

- Kernel function *k* encodes assumptions about correlation!

Covariance functions: Examples

- Linear kernel: $k(x, x') = x^T x'$

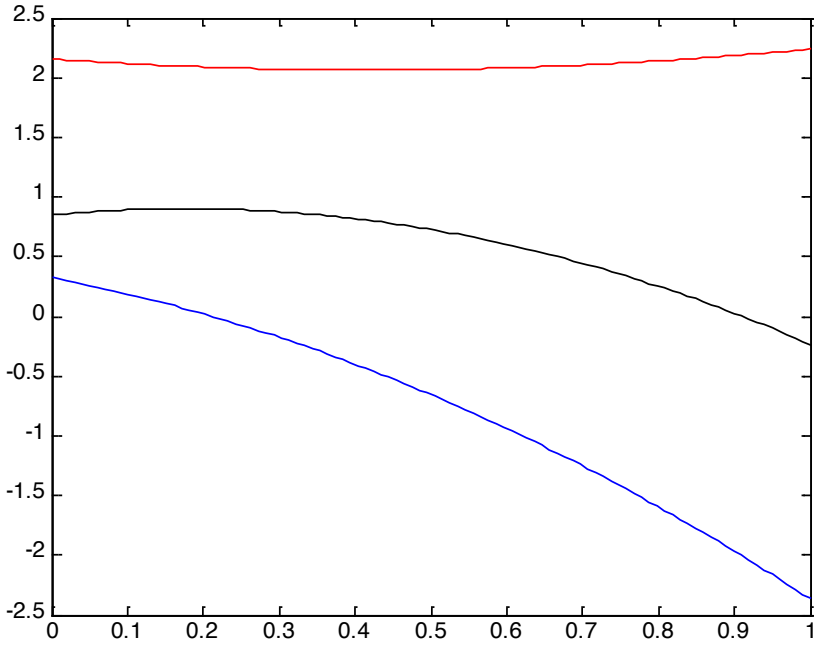


- GP with linear kernel = Bayesian linear regression

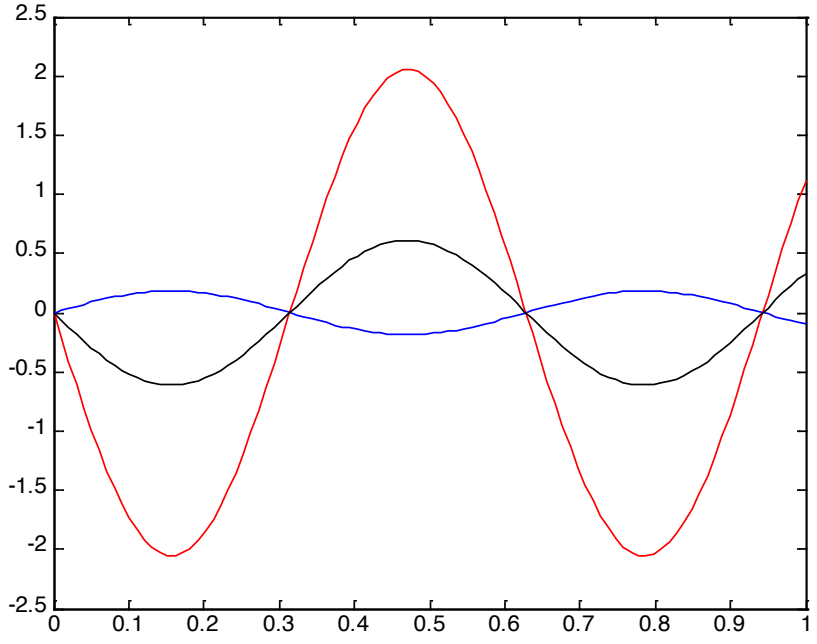
Covariance functions: Examples

- Linear kernel with features: $k(x, x') = \Phi(x)^T \Phi(x')$

E.g., $\Phi(x) = [1, x, x^2]$



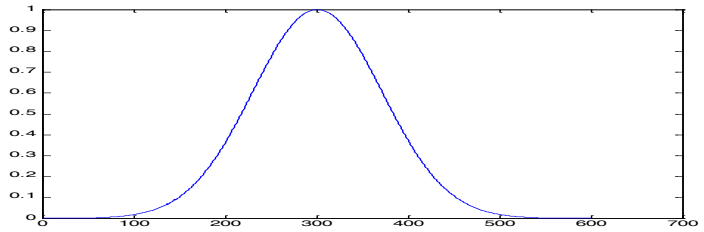
E.g., $\Phi(x) = \sin(x)$



Covariance functions: Examples

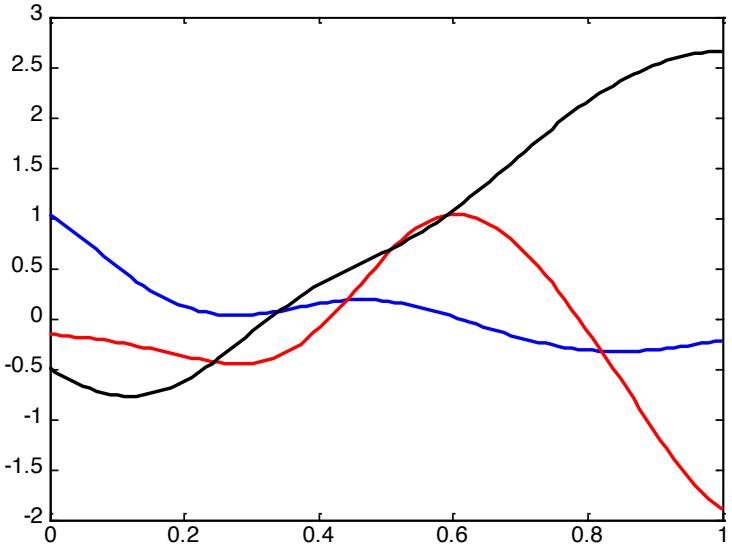
- Squared exponential (aka RBF, Gaussian) kernel

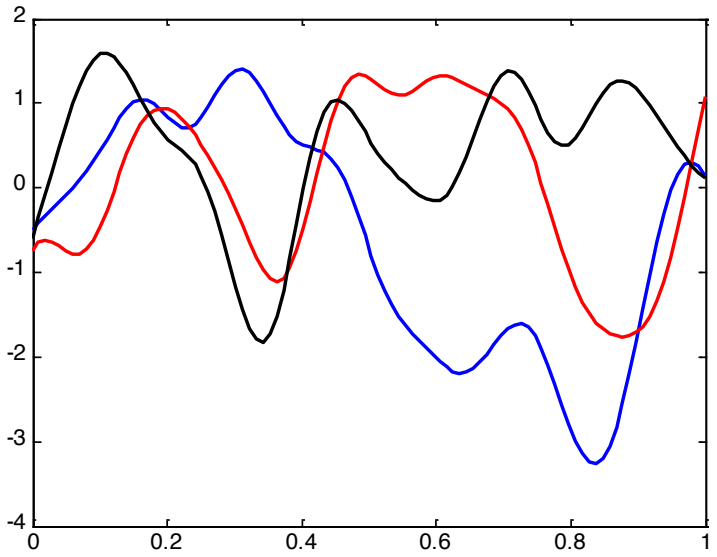
$$k(x, x') = \sigma_p^2 \exp\left(-\frac{\|x - x'\|_2^2}{2h^2}\right)$$



Distance $||x - x'||_2$

Samples from $P(f)$





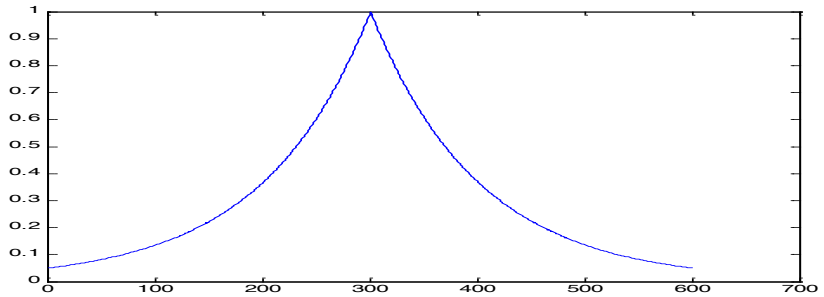
Bandwidth h=.3

Bandwidth h=.1

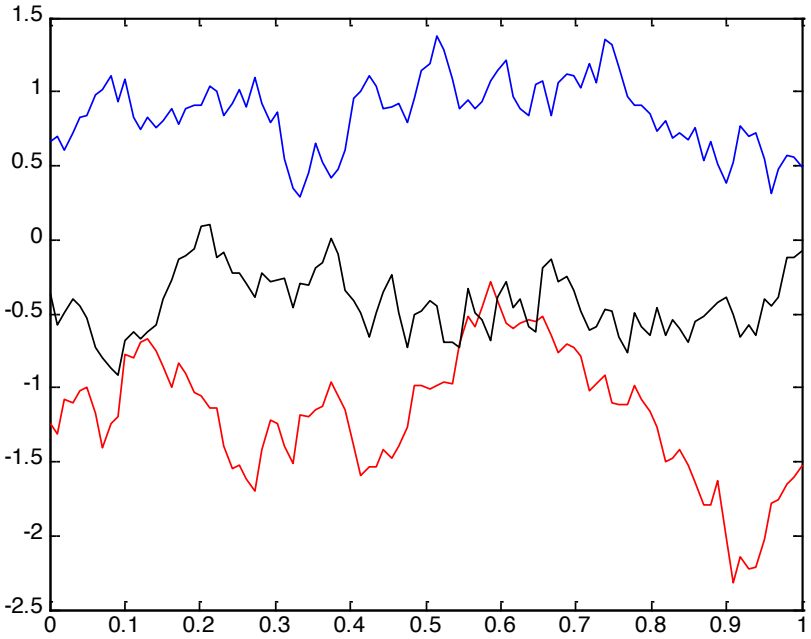
Covariance functions: Examples

- Exponential kernel

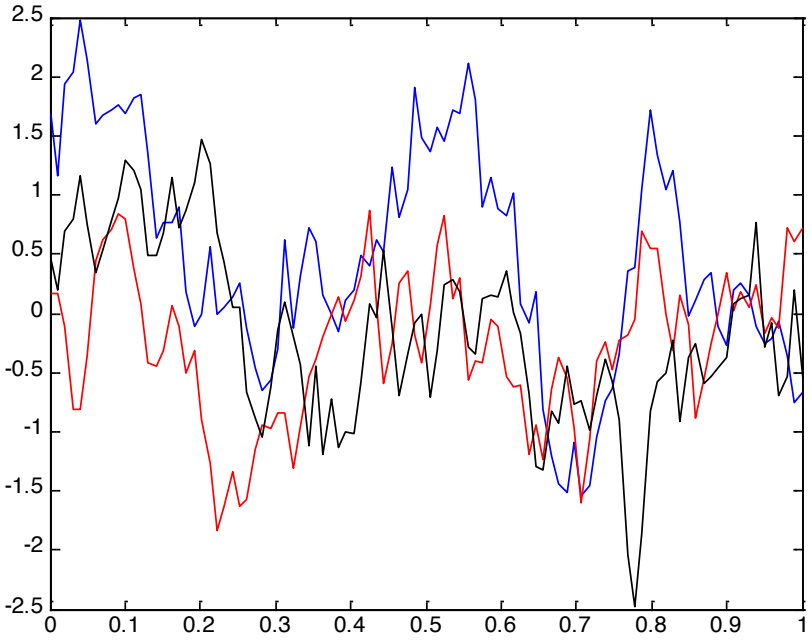
$$k(x, x') = \sigma_p^2 \exp\left(-\frac{\|x - x'\|_2}{h}\right)$$



Distance $||x - x'||_2$



Bandwidth $h = 1$



Bandwidth $h = .3$

Smoothness of GP samples

- Covariance function determines smoothness of sample paths*
  - Squared exponential kernel: analytic (infinitely diff'able)
  - Exponential kernel: continuous, but nowhere diff'able
  - Matérn kernel with parameter ν: $\lceil \nu \rceil - 1$ times (m.s.) diff'able

$$k(\mathbf{x}, \mathbf{x}') = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}||\mathbf{x} - \mathbf{x}'||_2}{\rho} \right)^{\nu} K_{\nu} \left(\frac{\sqrt{2\nu}||\mathbf{x} - \mathbf{x}'||_2}{\rho} \right)$$

Hereby Γ is the Gamma function, K_{ν} the modified Bessel function of the second kind, and ρ is a bandwidth param.

- Special cases: $\nu = \frac{1}{2}$ yields exponential kernel;
  $\nu \to \infty$ yields Gaussian kernel

Composition rules

- Suppose we have two covariance functions

$$k_1 : \mathcal{X} \times \mathcal{X} \to \mathbb{R} \qquad k_2 : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$$

  defined on data space *X*

- Then the following functions are valid cov. functions:

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}')\, k_2(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = c\, k_1(\mathbf{x}, \mathbf{x}') \text{ for } c > 0$$

$$k(\mathbf{x}, \mathbf{x}') = f(k_1(\mathbf{x}, \mathbf{x}'))$$

$f_1 \sim GP(0, k_1)$

$f_2 \sim GP(0, k_2)$

$f := f_1 + f_2 \Rightarrow f(x) = f_1(x) + f_2(x)$

$f \sim GP(0, k_1 + k_2)$

where *f* is a polynomial with positive coefficients or the exponential function

Forms of covariance functions

- Covariance function $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is called
  - Stationary if $k(x, x') = \tilde{k}(x - x')$
  - Isotropic if $k(x, x') = \tilde{k}(\|x - x'\|_2)$

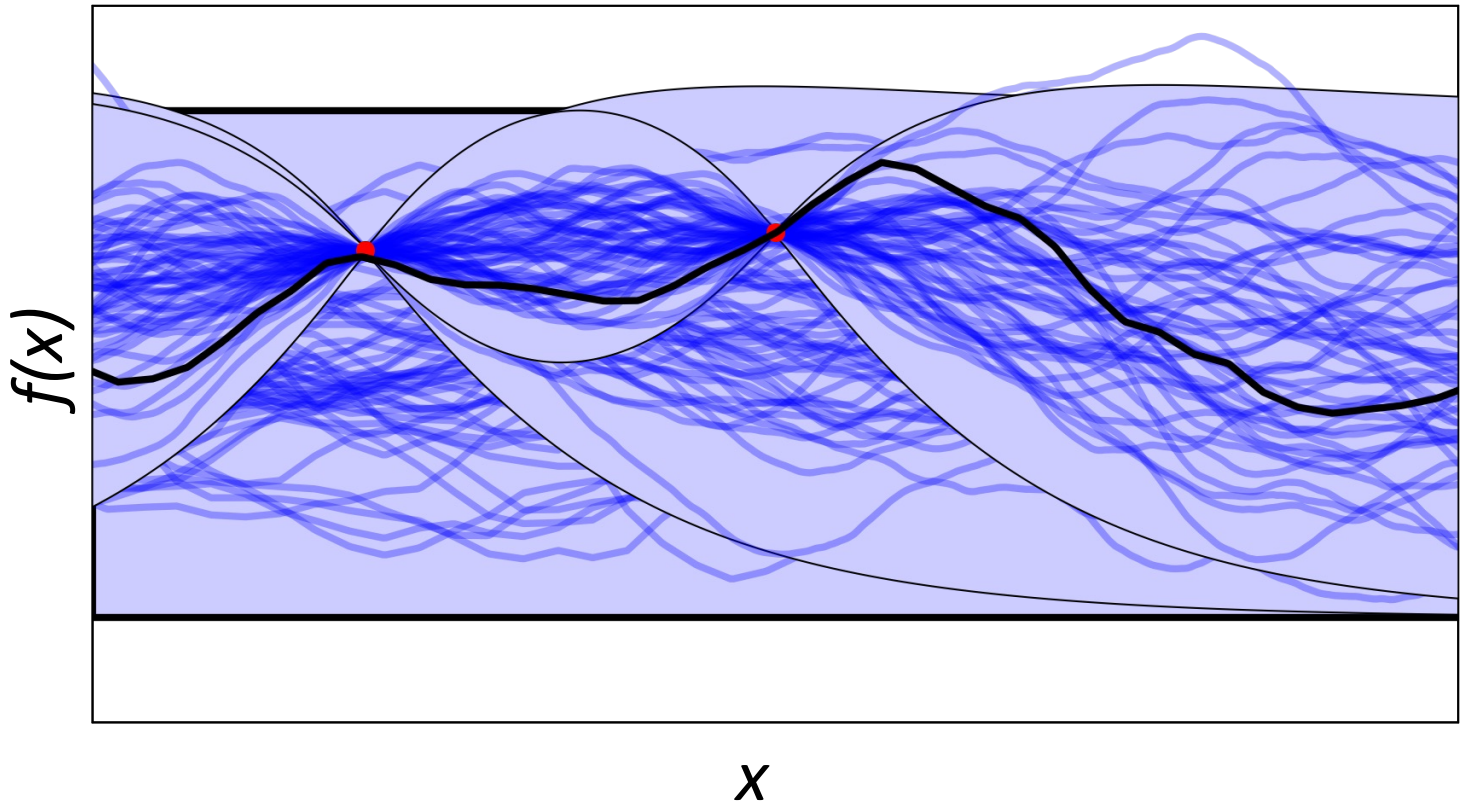
$\|x - x'\|_2^2 = (x - x')^T (x - x')$

Isotropic ⇒ stationary

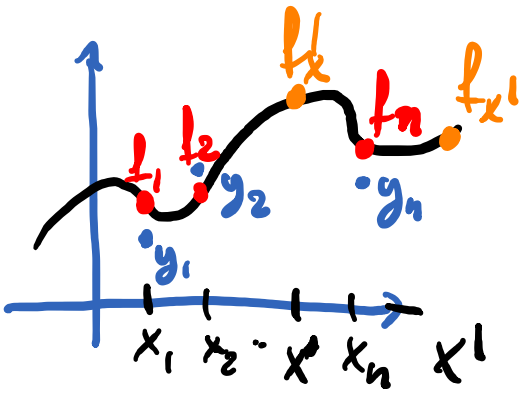
	Stationary?	Isotropic?
Linear $k(x, x') = x^T x'$	×	×
Gaussian $k(x, x') = \exp\left(-\frac{\|x - x'\|_2^2}{2h^2}\right)$	✓	✓
$k(x, x') = \exp\left(-(x - x')^T M (x - x')\right)$	✓	×

is a valid kernel ⇔ M pos. semidefinite

GP Inference Illustration



Making predictions with GPs



$$A = \{x_{1:n}, x, x'\}$$

$$y_i = f_i + \varepsilon_i \quad , \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$$

$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_n \\ f_x \\ f_{x'} \end{pmatrix} \sim \mathcal{N}(0, K_{AA})$$

$$z = \begin{pmatrix} y_1 \\ \vdots \\ y_n \\ f_x \\ f_{x'} \end{pmatrix} = f + \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \\ 0 \\ 0 \end{pmatrix} \sim \mathcal{N}\left(0, K_{AA} + \sigma_n^2 \cdot \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & 0 & \\ & & & & 0 \end{pmatrix}\right)$$

$$p(f_x, f_{x'} | y_{1:n}) = \mathcal{N}(\dots)$$

Making predictions with GPs

- Suppose $f \sim GP(\mu, k)$ and for set $A = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$

  we observe $y_i = f(\boldsymbol{x}_i) + \epsilon_i$ where $\epsilon_i \sim N(0, \sigma_n^2)$
- Then $p(f \mid \mathbf{x}_1, \ldots, \mathbf{x}_m, y_1, \ldots, y_m) = GP(f; \mu', k')$

  where

$$\mu'(x) = \mu(x) + \boldsymbol{k}_{x,A}(\boldsymbol{K}_{AA} + \sigma_n^2 \boldsymbol{I})^{-1}(\boldsymbol{y}_A - \mu_A)$$

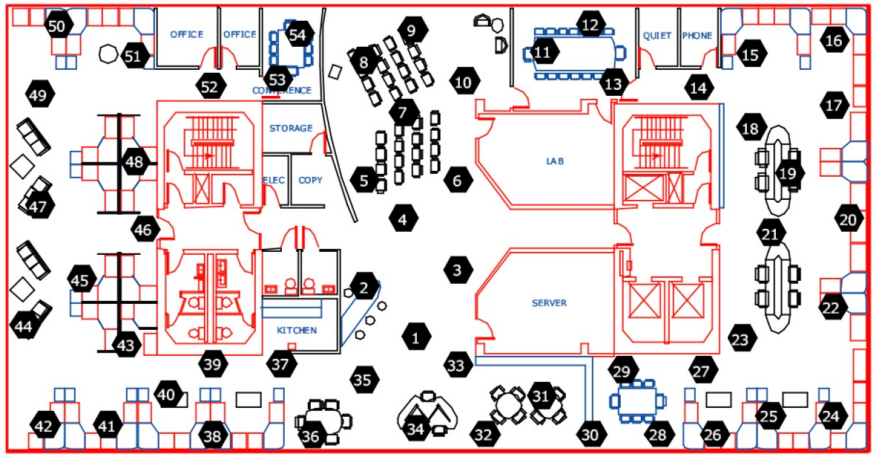
$$k'(x, x') = k(x, x') - k_{x,A}(K_{AA} + \sigma_n^2 I)^{-1} k_{x',A}^T$$

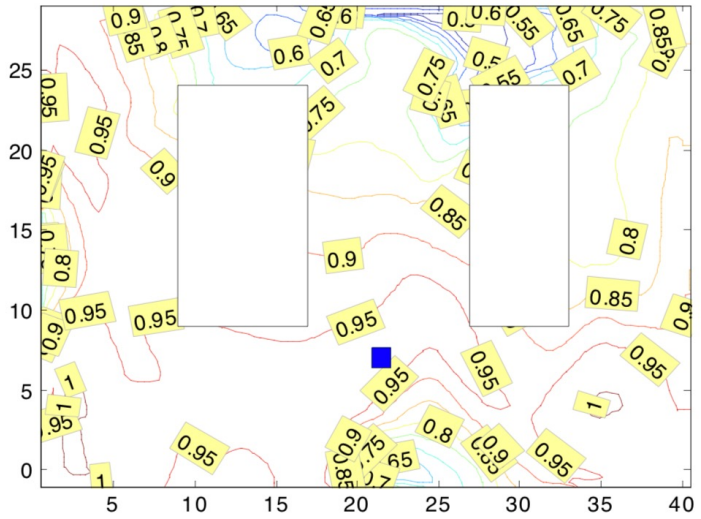
$k_{x,A} = [k(x, x_1) \ldots k(x, x_m)]$

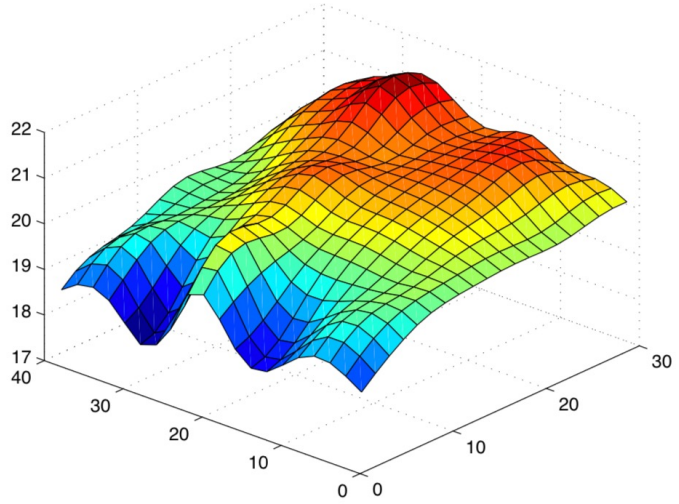
➔ Closed form formulas for prediction!

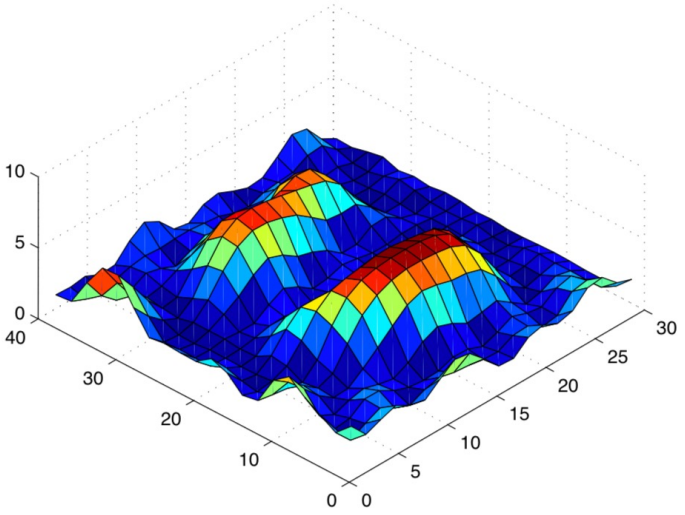
➔ Posterior covariance k' does not depend on $\boldsymbol{y}_A$

Application: Spatial prediction





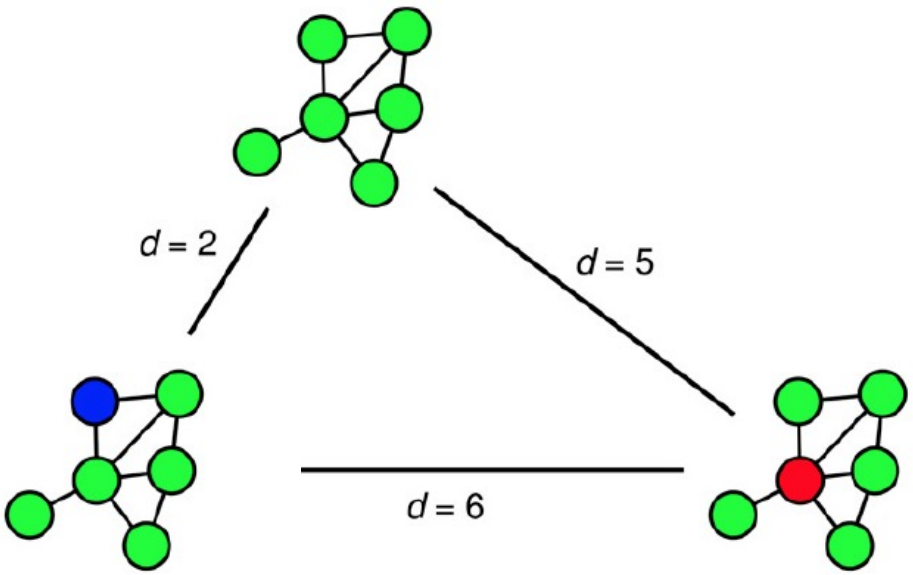


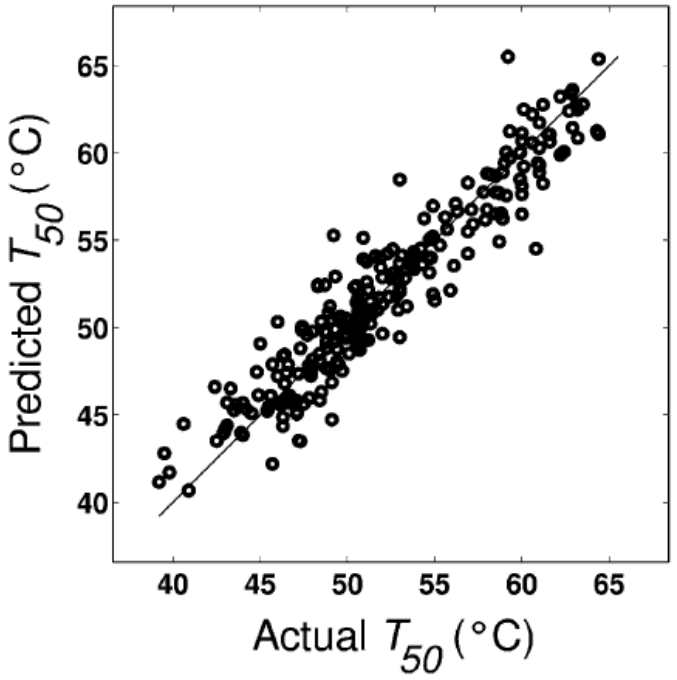


[Krause & Guestrin, Near-optimal Sensor Placements in Gaussian Processes, JMLR 2008]

Application: Protein Engineering

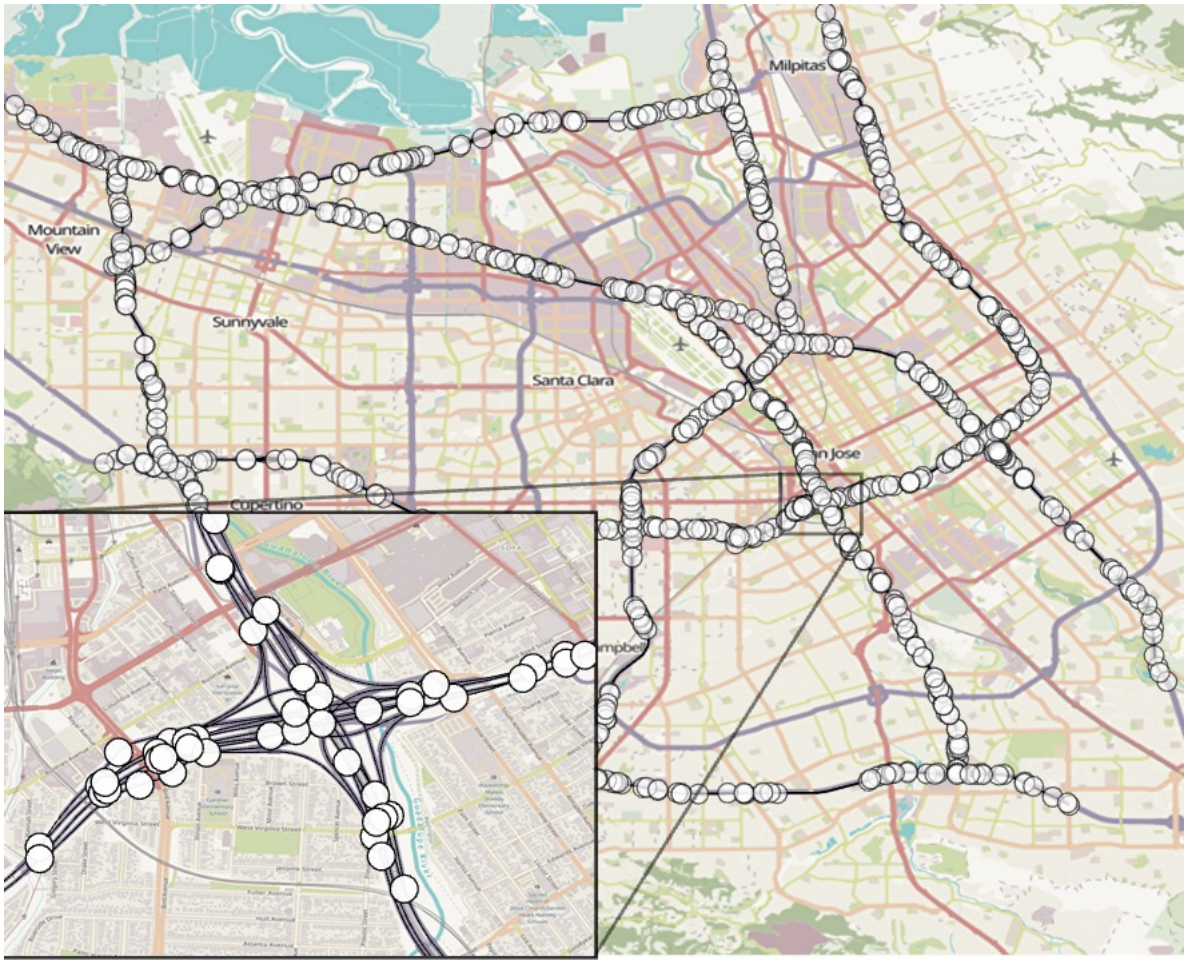
[with Romero, Arnold, PNAS '13]





Outlook: GPs on $X \neq \mathbb{R}^n$

Consider, e.g., X = node set of a road network graph.



San Jose highway network graph [Borovitskiy et al. AISTATS 2021]

Outlook: GPs on $X \neq \mathbb{R}^n$

Consider, e.g., X = node set of a road network graph.

Solution #1. Can embed X into $\mathbb{R}^2$ with

$$\xi(x) = (latitude(x), longitude(x))$$

and use $k(x, x') = \tilde{k}(\|\xi(x) - \xi(x')\|)$ with $\tilde{k}: \mathbb{R} \to \mathbb{R}$.

Problem: travel distance can be large even if the nodes are close to each other in terms of $\|\xi(x) - \xi(x')\|$.

E.g., opposite moving directions in the same location are far from each other.

A traffic jam may be only one side.



Outlook: GPs on $X \neq \mathbb{R}^n$

Consider, e.g., X = node set of a road network graph.

Solution #2. Use $k(x, x') = \tilde{k}\big(d(x, x')\big)$ with $\tilde{k}: \mathbb{R} \to \mathbb{R}$ and $d(x, x')$ being the shortest path distance on the graph.

Problem: k will typically fail to be positive semi-definite.

E.g., $\tilde{k}\big(d(x, x')\big) = e^{-\frac{d(x,x')^2}{h^2}}$ is positive semi-definite for all $h > 0$ if and only if X can be isometrically embedded into $\mathbb{R}^n$ for high enough n or into an infinite-dim Hilbert space.

[Schoenberg Trans. Am. Math. Soc. 1938]

Outlook: GPs on $X \neq \mathbb{R}^n$

Consider, e.g., X = node set of a road network graph.

Solution #3. *Laplacian*-based generalizations of the squared exponential and Matérn kernels.

Properties:

☺ Always positive semi-definite.

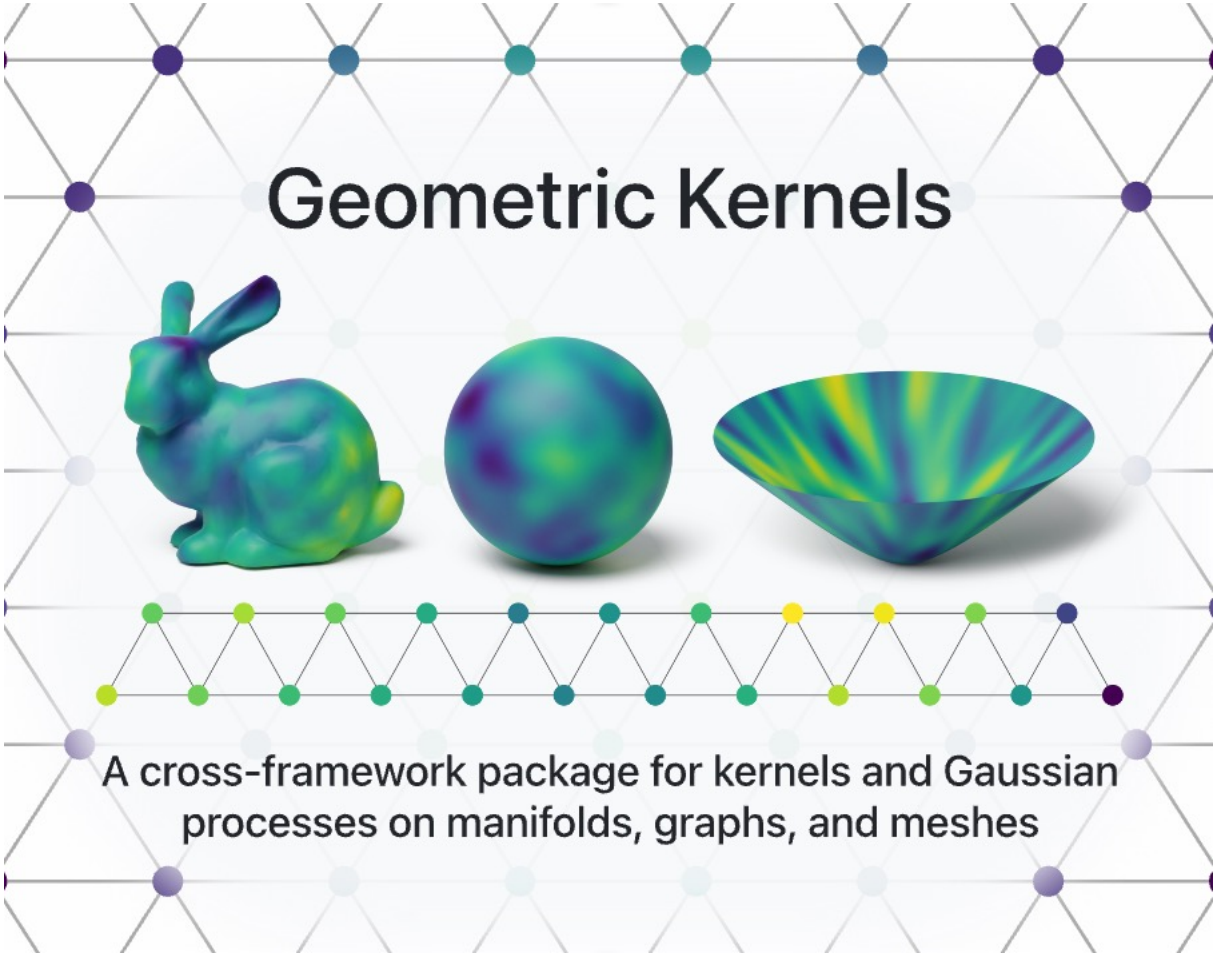
☺ Automatically respect the intrinsic geometry of X.

☹ If $|X| > 10^5$, requires non-trivial tricks/compute.

[Kondor & Lafferty ICML 2002, Borovitskiy et al. AISTATS 2021]

Outlook: GPs on $X \neq \mathbb{R}^n$

Same is possible for other X: manifolds, meshes, etc.



https://geometric-kernels.github.io/

Common convention: Prior mean 0

w.l.o.g.:

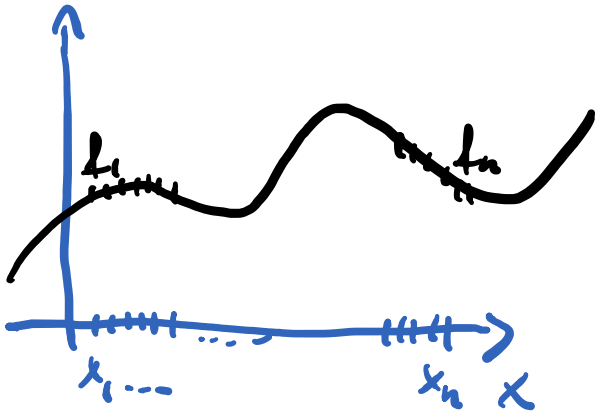
Sps $f \sim GP(\mu, k)$

$g := g(x) := f(x) - \mu(x)$

$\rightsquigarrow g \sim GP(0, k)$

$\hookrightarrow f(x) = g(x) + \mu(x)$

How to sample from a GP?



$f = [f_{1:n}] \sim \mathcal{N}(0, K)$

$K = LL^T$ Cholesky-decomp.

Sample $\varepsilon \sim \mathcal{N}(0, I) \Rightarrow \varepsilon_i \overset{iid.}{\sim} \mathcal{N}(0,1)$

$f := L \cdot \varepsilon$

$x_1 \dots x_n, \dots$

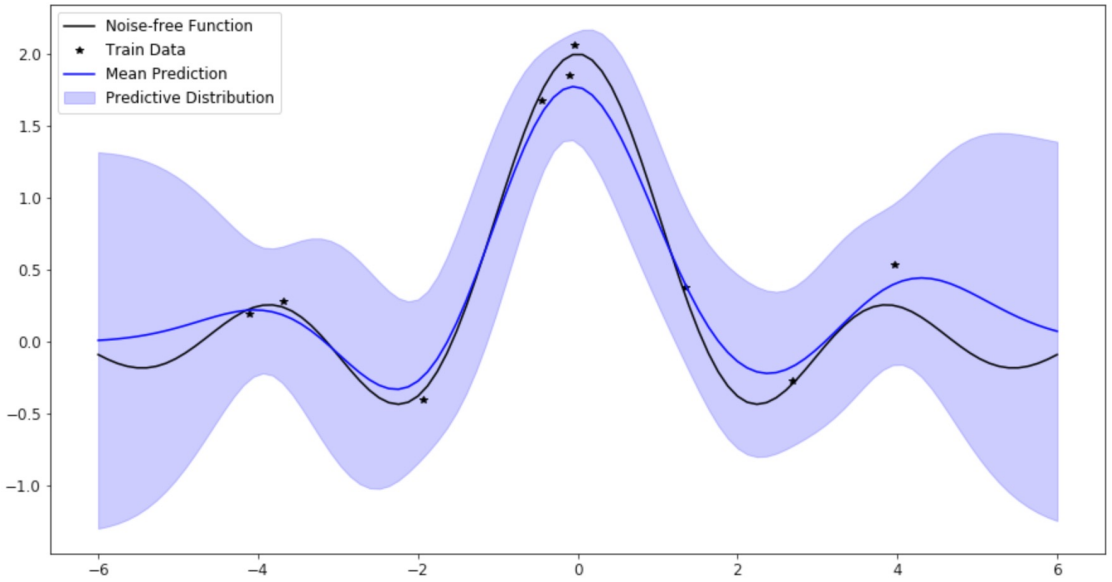
$p(f_1, \dots f_n, \dots)$

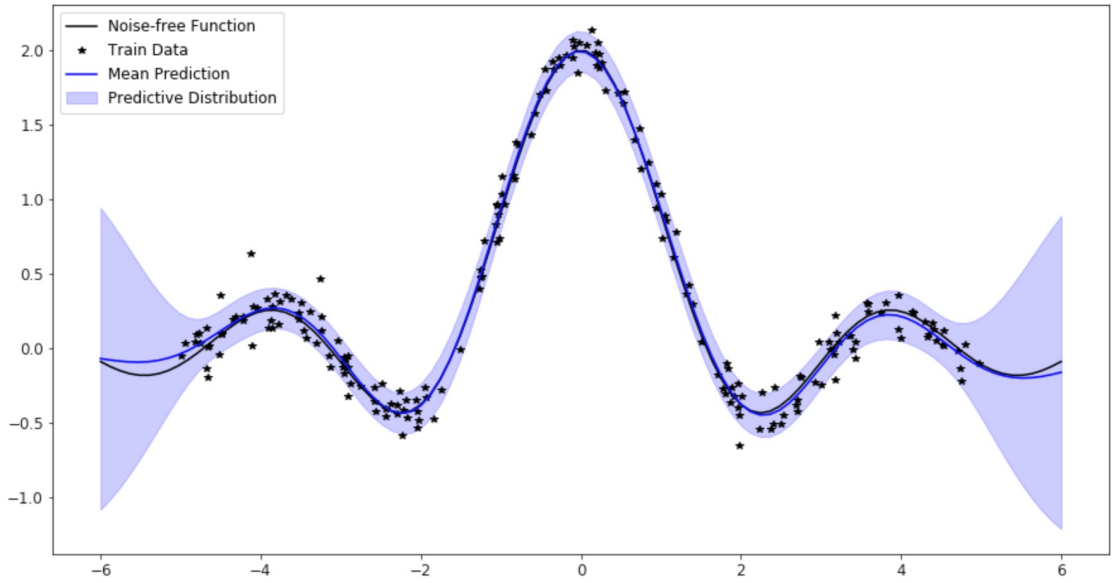
$= p(f_1) \cdot p(f_2 | f_1) \cdots p(f_n | f_{1:n-1})$

$\cdots$

"forward sampling"

Demo


Noise-free Function
Train Data
Mean Prediction
Predictive Distribution
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−6
−4
−2
0
2
4
6


Noise-free Function
Train Data
Mean Prediction
Predictive Distribution
2.0
1.5
1.0
0.5
0.0
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0
2
4
6